SQL250

Transact-SQL for Developers

MOC: DP-080T00 | 5 Days | Labs | Live Q&A

In this course, you'll establish a solid foundational understanding of database concepts and terminology. Then you'll master the use of various Microsoft tools to submit queries quickly and efficiently get the results you want.

We teach students how to write a query the same way that SQL Server processes a SQL statement – following a step-by-step process for creating SQL queries from business requirements. This approach uses the natural way of breaking down a problem into logical steps. Each step can be validated before moving to the next step. This differs from most courses which present SQL as a set of features.

Finally, you'll learn to harness the power of using T-SQL statements inside common database objects like Views, Stored Procedures and User-Defined Functions.

SQL250 is a superset of SQL101: Introduction to Transact-SQL. It covers everything in SQL101 but goes more in-depth, moves at a faster pace, and adds a full module on T-SQL development – stored procedures, user-defined functions, views, and more. Designed for students who have programming experience in any language.

Class lectures highlight and explain Transact-SQL concepts which are reinforced with extensive follow-along demonstrations and hands-on labs.

What You Will Learn

- Understand relational database concepts, table structures, and how data is organized in SQL Server
- Navigate SQL Server Management Studio (SSMS) to discover database objects, submit queries, and view results
- Write SELECT statements to retrieve data from one or more tables using inner joins, outer joins, and self-joins
- Combine data from multiple tables using various join types and derived-table subqueries
- Filter query results using the WHERE clause with comparison operators, logical operators, BETWEEN, IN, LIKE, and NULL handling
- Understand data type precedence and implicit conversions and their impact on query results
- Control query output with the SELECT clause, column aliases, expressions, and the ORDER BY clause
- Use built-in system functions for string manipulation, date/time operations, mathematical calculations, and type conversions
- Apply the CASE expression for conditional logic within queries
- Aggregate data using GROUP BY with aggregate functions (COUNT, SUM, AVG, MIN, MAX) and filter groups with HAVING
- Rank and partition data using window functions (ROW_NUMBER, RANK, DENSE_RANK, NTILE)
- Perform data maintenance operations: INSERT, UPDATE, DELETE, and MERGE

- Manage transactions to ensure data integrity with BEGIN TRANSACTION, COMMIT, and ROLLBACK
- Create and use Views to encapsulate complex queries and simplify data access
- Write Scalar User-Defined Functions for reusable calculations and logic
- Build Inline Table-Valued Functions and Multi-Statement Table-Valued Functions
- Develop Stored Procedures with input/output parameters, conditional logic, and error handling
- Apply basic programming constructs (variables, IF/ELSE, WHILE loops) within T-SQL scripts

Course Outline

Module 1: The Basics

The material we cover in this module will be used throughout the course. Understanding objects and object names, the processing order of a query, literal values, identifiers and delimiters, SSMS basics, and more – these are things that you will use no matter what you do in SQL.

Lessons

- Introducing databases
- The database table
- SQL basics
- SSMS basics
- Query basics
- The metadata

Lab

- Database Concepts and SSMS

Module 2: The Relational Model

Any discussion about relational databases will inevitably require an understanding of the Relational Model. The RM describes the organization and structure of the database and will have an impact on almost everything in this course and back at work. Also, understanding this important topic is the first step in understanding and applying joins.

Lessons

- Introducing the Relational Model
- The need for joins
- Table relationships
- OLAP and OLTP databases (Optional – Time permitting)

Lab

- Table Relationships

Module 3: Getting the Data

The FROM clause always references a table. The table can be a database table, it can be generated by table-type functions, views, subqueries or joins. In this module, our focus will be on joins and subqueries.

Lessons

- The FROM clause
- Single-join queries
- Multiple-join queries
- Derived-table Subqueries
- Old join syntax

Lab

- Combining two tables with a Join
- Combining multiple tables with a Join

Module 4: Filtering the Data

Now that we can use the FROM clause to define the data for our query, we need a way to select what we want to include in our results and what we want to exclude. In this module, we look at different ways we can use the WHERE clause to filter the data.

Lessons

- Introduction
- Expressions
- Data type precedence
- Advanced filtering
- WHERE clause optimization

Lab

- Defining the result set

Module 5: Displaying the Data

At this point, we have learned how to define the data to query using the FROM and WHERE clauses. We are ready to display the results from our query. This is where we project columns and apply any sorting of the results. We will also take advantage of this opportunity to dive a little more deeply into functions.

Lessons

- Introduction
- The column list
- The ORDER BY Clause
- Functions
- CASE expression

Lab

- Controlling output rows

Module 6: Aggregating the Data

All levels of management need information to provide decision support. Lower levels of management may need to make decisions about what products and how much to have in stock, and higher levels of management may need to understand industry trends to support strategic decisions about where to guide the company. In this module, you will learn how to create Aggregate queries; they provide this powerful type of high-level information.

Lessons

- Introduction
- Aggregation with GROUP BY
- Advanced aggregation
- Ranking data

Lab

- Grouping data

Module 7: Maintaining the Data

Everything we've seen so far in this course has been about reading the data. In this module, we look at the three write operations that we collectively call data maintenance: Insert, delete, and update. We start with a look at transactions, and we end with a look at the Merge command that allows us to perform all three data maintenance operations in one pass of the data.

Lessons

- Transactions
- Inserting data
- Deleting data
- Updating data
- Merging data

Lab

- Data maintenance operations

Module 8: Development in T-SQL

So far in this course, we've seen how the conceptual schema works. In this module we turn to the external schema where we can encapsulate much of what we've learned. We will encapsulate our age calculation, see how a stored procedure can be used to save a 1:N relationship, process transactions, perform data validation, and more.

Lessons

- Basic Programming constructs
- Scalar functions
- Views
- Inline table-valued functions
- Stored procedures | Part 1
- Stored Procedures | Part 2
- Multi-statement table-valued functions

Lab

- Final scenario

Audience

This course is intended for SQL Server Developers, Database Administrators, and System Engineers who are responsible for writing T-SQL queries for an application. It is also suitable for Business Analysts and other professionals who need to query SQL Server databases and have some programming background.

Prerequisites

Before attending this course, students should have:

- Familiarity with programming and developing software using any language

SQL250 vs. SQL101

	SQL101	SQL250
Target Audience	No programming experience	Has programming experience
Pace	Standard	Accelerated
Modules 1–7	Core T-SQL query skills	Same content, greater depth
Module 8: T-SQL Development	—	Views, Stored Procedures, UDFs, Programming constructs
WHERE Optimization	—	Included
Old Join Syntax	—	Included
Advanced Aggregation	—	Included
Ranking Functions	—	Included
Transactions & MERGE	—	Included
Duration	5 Days	5 Days
MOC Alignment	—	DP-080T00